

Diffusion-2BC: Hybrid Diffusion and Regression Training for Offline Behavior Cloning in Autonomous Driving

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Abstract

Behavior cloning provides an offline route to autonomous-driving policy learning, but mean-squared-error regression is poorly matched to demonstrations in which one observation admits several valid actions. Diffusion policies can represent conditional multimodal action distributions, yet their closed-loop performance may be unstable when visual features and control are learned from limited data. This paper presents Diffusion-2BC, which combines a diffusion denoising objective with an auxiliary deterministic behavior-cloning loss over a shared visual encoder. The auxiliary branch is used only during training; inference remains diffusion-based. The proposed method is evaluated in the controlled Claw environment and in bird’s-eye-view CARLA navigation, including route-conditioned driving, route-free navigation through multiple intersections, and cross-map evaluation from Town01 to Town02. In the Claw task, Diffusion-2BC reduced the mean mask-distance error by approximately 10% relative to a diffusion-based behavior-cloning baseline and by 85% relative to standard deterministic behavior cloning. In route-free CARLA, Diffusion-2BC traveled substantially farther before termination under the evaluation protocol than both baselines in Town01 and Town02. Additional qualitative rollouts revealed distinct route choices, showing the multimodal behavior of the proposed diffusion-based agent. The results indicate that an auxiliary regression signal can improve the closed-loop reliability of diffusion behavior cloning while preserving multimodal prediction in the controlled benchmark.

Keywords: autonomous driving; behavior cloning; diffusion models; imitation learning; multimodal policies; CARLA

1 Introduction

Autonomous driving requires perception, decision-making, and control under uncertainty. A central difficulty is that a driving situation does not always imply one uniquely correct action. At an intersection, a driver may turn left or right; when approaching a bend, different combinations of steering and acceleration may remain safe; and when slowing down, drivers may adopt smoother or more abrupt profiles. Such alternatives are not necessarily annotation errors. They can be valid modes of an expert’s behavior and should be represented by a policy intended to imitate diverse demonstrations.

Imitation learning is attractive in this setting because it learns from demonstrations rather than from a manually engineered reward. Behavior cloning (BC), the most direct imitation-learning formulation, treats policy learning as supervised prediction from observations to actions [1, 2]. Its simplicity makes it computationally efficient and fully compatible with fixed offline datasets. Nevertheless, two limitations are important in closed-loop driving. First, errors compound because the learned policy encounters states that were not represented in the expert data [3, 4]. Second, the usual mean-squared-error (MSE) objective represents the conditional mean of the action distribution. When demon-

strations contain separated modes, that mean may correspond to an unsafe action or the network may collapse to a single dominant mode [5, 6].

Generative policies offer a way to model a distribution rather than a point estimate. Diffusion models learn to reverse a progressive noising process and have been effective in high-dimensional generation [7, 8, 9]. Recent work has adapted them to planning, decision-making, and visuomotor control [10, 11, 12, 13]. In behavior cloning, the reverse diffusion process can generate different actions for similar observations, which directly addresses multimodality without discretizing continuous controls.

Expressiveness alone, however, does not guarantee reliable closed-loop driving. A diffusion policy must learn useful visual features and a denoising process from a finite expert dataset, and every action requires multiple network evaluations. Our experiments indicated that a basic diffusion behavior-cloning policy could express conflicting left/right decisions but could also be inconsistent in complex route-free urban navigation. This motivates the central question of this paper: *can a direct deterministic behavior-cloning signal improve the representation learning and closed-loop performance of a diffusion policy without eliminating multimodal action generation?*

We answer this question with Diffusion-2BC, a hybrid training architecture. A convolutional feature extractor feeds both a diffusion denoising branch and an auxiliary fully connected action-regression branch. Their losses are combined by a coefficient α . The auxiliary branch is present only during training, so the test-time policy remains stochastic and uses the same reverse process as standard diffusion behavior cloning. The method is evaluated entirely offline: no environment interaction is used to update the policy after the demonstration datasets have been collected.

The evaluation in this article follows a staged experimental argument. The controlled Claw environment provides a test of multimodal conditional prediction, whereas CARLA provides the main closed-loop evaluation of route following, route-free navigation, qualitative route diversity, and cross-map behavior. This organization clarifies which conclusions are supported by each environment and avoids using preliminary experiments as evidence for the proposed method.

1.1 Relation to preliminary work and CarRacing motivation

A preliminary conference paper evaluated standard diffusion behavior cloning against deterministic behavior cloning in CarRacing and CARLA, including a controlled T-intersection experiment [14]. The present journal article substantially extends that study by introducing the Diffusion-2BC architecture and hybrid objective, evaluating the proposed method in route-conditioned and route-free multi-intersection CARLA navigation, adding the Town01-to-Town02 cross-map analysis, and expanding the methodological, computational-cost, failure-mode, and validity discussions. The controlled T-intersection experiment is not repeated because it compares only MSE-BC and Diffusion-BC and therefore does not evaluate the proposed method.

Before developing Diffusion-2BC, the conference study also compared deterministic MSE-BC and Diffusion-BC in the visually observed CarRacing environment [14]. Two independently trained models of each architecture were evaluated on 100 randomly generated tracks per model. Diffusion-BC achieved a higher aggregate mean reward and lower variability (861.35 ± 48.07) than MSE-BC (638.79 ± 105.20), indicating that diffusion-based action generation could improve visual closed-loop control. This previously published result is summarized here as empirical motivation for adding a direct regression signal to diffusion training. Supplementary Note S1 summarizes the protocol, reproduces the previously published per-episode reward distributions for context, and adds cumulative-reward curves that were not included in the conference article.

The contributions are as follows:

- We formulate Diffusion-2BC, an offline behavior-cloning architecture that combines denoising and deterministic regression objectives in a shared visual representation while preserving diffusion-only inference.

- We compare MSE-BC, Diffusion-BC, and Diffusion-2BC in the controlled Claw benchmark and in route-conditioned and route-free CARLA navigation, including cross-map evaluation from Town01 to Town02.
- We analyze the trade-off among route progress, behavioral diversity, computational cost, and failure modes, and delimit the conclusions supported by the evaluated runs and baselines.

2 Related Work and Positioning

2.1 Behavior cloning for autonomous driving

Early end-to-end driving systems already used supervised imitation to map visual input to steering commands [1]. Modern behavior-cloning systems use deeper visual encoders and richer sensor representations, but the central supervised formulation remains unchanged. Conditional imitation learning addresses route ambiguity by providing a high-level command such as turn left, turn right, or continue straight [15]. Such conditioning is effective when command labels are available, but it does not solve the distinct problem considered here: learning the distribution of unlabeled valid actions that are jointly present in the demonstrations.

Driving-policy errors also arise from distribution shift. Small mistakes move the vehicle away from expert states, after which predictions can degrade rapidly. Dataset aggregation and robustification methods reduce this problem by collecting additional states or synthesizing perturbations [3, 4, 16]. The present study does not use online relabeling or additional recovery demonstrations; it asks how far an offline policy can progress using the available fixed datasets.

Learning-based decision and control methods have also been investigated in the intelligent-transportation literature for autonomous vehicles operating at intersections and under changing environmental conditions. Emamifar and Ghoreishi [17] proposed a physics-informed reinforcement-learning framework for autonomous-vehicle control at signalized intersections, combining uncertain system dynamics with partial sensor information. More recently, Wang and Ghoreishi [18] combined pre-trained reinforcement-learning policies with probabilistic reasoning to support robust decision making under non-stationary conditions, with evaluation in CARLA. These approaches address robustness and decision making through reinforcement-learning-based formulations, whereas the present work focuses on fully offline imitation from fixed demonstrations and on representing multimodal action choices without online policy adaptation.

2.2 Multimodal imitation learning

Several approaches have been proposed for multimodal action distributions. Discretization converts continuous control into classification but introduces quantization and can separate dimensions that should remain coordinated. Mixture-density

and latent-variable models represent several modes explicitly, while energy-based methods score candidate actions without requiring a normalized output distribution. Implicit Behavioral Cloning, for example, models the observation-action compatibility through an energy function [5]. Behavior Transformers represent multiple action modes using discretized latent structure [6]. These methods demonstrate that a unimodal regression loss is not the only viable BC objective.

The evaluation compares a deterministic BC baseline, a diffusion baseline, and the proposed hybrid objective. It therefore establishes the effect of the auxiliary regression signal relative to the implemented baselines, rather than superiority over every multimodal imitation-learning family.

2.3 Diffusion models for decision-making and control

Diffusion models define a forward Markov chain that progressively corrupts data and a learned reverse chain that reconstructs samples [8]. Conditional diffusion extends this mechanism by conditioning the denoising network on context. In decision-making, diffusion models have been used to generate trajectories, action sequences, and control outputs [11, 19, 12]. Their iterative sampling supports multimodality but increases inference cost.

The closest baseline in this work, Diffusion-BC, conditions denoising on features extracted from the current observation and predicts the noise applied to an expert action. Diffusion-2BC adds an auxiliary deterministic head and MSE loss during training. This resembles the broader principle of supplementing a difficult imitation objective with a direct BC signal, as explored in BC-augmented adversarial imitation learning [20]. The distinction is that Diffusion-2BC remains fully offline, does not train a discriminator, and removes the deterministic branch at inference.

Diffusion Model-Augmented Behavioral Cloning also combines a BC objective with a diffusion-model objective, but it learns a conventional conditional policy together with a diffusion model of the joint expert state-action distribution [21]. In contrast, Diffusion-2BC uses the conditional diffusion model itself as the executed policy and employs deterministic regression only as an auxiliary training signal. Thus, the methods share the idea of complementary conditional and diffusion losses but assign the diffusion component different roles at inference.

3 Background

3.1 Offline behavior cloning

Let $\mathcal{D} = \{(o_i, a_i)\}_{i=1}^N$ denote a fixed dataset of expert observation-action pairs. The deterministic baseline learns a policy π_θ by minimizing

$$\mathcal{L}_{\text{MSE}} = \mathbb{E}_{(o,a) \sim \mathcal{D}} [\|\pi_\theta(o) - a\|_2^2]. \quad (1)$$

For a conditional action distribution with one concentrated mode, this objective is appropriate. For separated modes, its minimizer is the conditional mean, which need not coincide with a demonstrated action. In practice, a finite neural network may instead converge to one mode. Both outcomes are undesirable when the goal is to reproduce diverse valid strategies.

3.2 Forward diffusion process

Let a_0 be an expert action. A variance schedule $\{\beta_t\}_{t=1}^T$ defines $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. The forward process adds Gaussian noise:

$$q(a_t | a_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}a_{t-1}, \beta_t I). \quad (2)$$

Using the reparameterized closed form, a noisy action at any timestep can be sampled directly:

$$a_t = \sqrt{\bar{\alpha}_t}a_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (3)$$

As t increases, the action becomes progressively less distinguishable from Gaussian noise.

3.3 Conditional reverse process for behavior cloning

A denoising network ϵ_θ predicts the noise used in Eq. (3). It receives the noisy action a_t , the diffusion timestep t , and features $f_\theta(o)$ extracted from the current observation. The simplified training objective is

$$\mathcal{L}_{\text{DBC}} = \mathbb{E}_{o,a,t,\epsilon} [\|\epsilon_\theta(f_\theta(o), a_t, t) - \epsilon\|_2^2]. \quad (4)$$

At inference, the policy samples $a_T \sim \mathcal{N}(0, I)$ and repeatedly applies the learned reverse update until an action estimate a_0 is obtained. Different initial noise samples can produce different valid actions under the same observation, which is the mechanism used to express multimodality.

4 Proposed Diffusion-2BC Method

4.1 Design rationale

The experiments motivating the method showed complementary behavior. Deterministic BC was fast and could learn route following but failed to represent conflicting decisions at the same intersection. Diffusion BC represented left and right turns with a single model but was less consistent in the more difficult route-free, multi-intersection setting. We therefore retain diffusion inference and add a direct regression signal only during representation learning.

4.2 Architecture and objective

Fig. 1 shows the training architecture. The observation is first processed by a shared convolutional feature extractor f_θ . Its

Table 1 Qualitative positioning of the methods most directly related to this study. “Unlabeled multimodality” means that several valid actions may appear for similar observations without an explicit command identifying the desired mode

Method family	Action representation	Handles unlabeled multimodality	Main strength	Limitation relative to this study
MSE behavior cloning	Deterministic point estimate	No	Simple, fast offline training and inference	Conditional averaging or single-mode behavior
Command-conditioned BC	Deterministic action conditioned on a route command	Only after labeling modes	Direct controllability at intersections	Requires reliable command or route labels
Energy-based / implicit BC	Energy over observation-action pairs	Yes	Flexible multimodal scoring	Requires candidate optimization or sampling; not evaluated here
Behavior Transformer	Discrete latent/action representation	Yes	Represents several behavior modes	Quantization and architecture choices differ from continuous diffusion
Diffusion-augmented BC	Conditional BC policy plus joint state-action diffusion model	Potentially	Combines conditional and joint expert modeling	The direct policy, rather than the diffusion model, is the primary action predictor
Diffusion-BC	Conditional generative action model	Yes	Continuous stochastic action generation	Iterative inference and inconsistent closed-loop behavior in the hardest experiment
Diffusion-2BC (proposed)	Diffusion policy with auxiliary regression training signal	Yes	Preserves stochastic inference while improving shared features	Additional training complexity; not compared with all multimodal BC families

output conditions a denoising network and is also passed to an auxiliary fully connected head g_θ that predicts the expert action directly. The combined objective is

$$\mathcal{L}_{\text{D2BC}} = \alpha \mathcal{L}_{\text{MSE}} + (1 - \alpha) \mathcal{L}_{\text{DBC}}, \quad 0 \leq \alpha \leq 1, \quad (5)$$

where

$$\mathcal{L}_{\text{MSE}} = \|g_\theta(f_\theta(o)) - a\|_2^2. \quad (6)$$

The coefficient α can be fixed or scheduled. A fixed value of 0.3 is used in the route-conditioned CARLA experiment. In the multi-intersection experiment, α is updated once per epoch using $\alpha_e = \exp(-ke)$, with $k = -\ln(0.01)/E$, so that the schedule starts at 1.0 and decays toward 0.01 over E training epochs. The study does not include a systematic hyperparameter ablation, so these settings are treated as practical design choices rather than globally optimal values.

4.3 Network architecture

All evaluated policies use the same visual encoder. It comprises two residual convolutional blocks, each with two 3×3 convolutions (stride 1, padding 1), batch normalization, and GELU activations. The blocks use 64 and 128 output channels, respectively; their residual sums are scaled by $1/\sqrt{2}$, and each block is followed by 2×2 max pooling. An 8×8

average-pooling layer and flattening produce the visual representation. The input and final action projections are adapted to each environment.

The conditional denoising backbone follows the observation-to-action Diffusion-BC architecture of Pearce et al. [13]. The visual representation, noisy action, and normalized diffusion timestep are embedded separately into 128-dimensional vectors and projected to three 64-dimensional tokens. Four Transformer encoder blocks process these tokens, each using 16 attention heads, residual connections, batch normalization, and a feed-forward expansion from 64 to 256 units with GELU activation. The flattened Transformer output is projected to the action-noise dimension. No Transformer dropout is used in the reported CARLA runs, and classifier-free context dropout and guidance are disabled ($p_{\text{ctx}} = 0$, guidance weight = 0).

For CARLA, the convolutional encoder produces a 4,608-dimensional flattened representation. The deterministic MSE-BC baseline maps this vector through hidden widths 2,304, 1,024, and 64 to the two-dimensional control output. Diffusion-2BC retains the same denoising branch as Diffusion-BC and adds an auxiliary regression head $4608 \rightarrow 2048 \rightarrow 512 \rightarrow 128 \rightarrow 32 \rightarrow 2$, with Batch-Norm, ReLU, and dropout $p = 0.3$ after each hidden layer. The auxiliary head is used only during training. Table 2 summarizes the resulting policy structures, while Supplementary

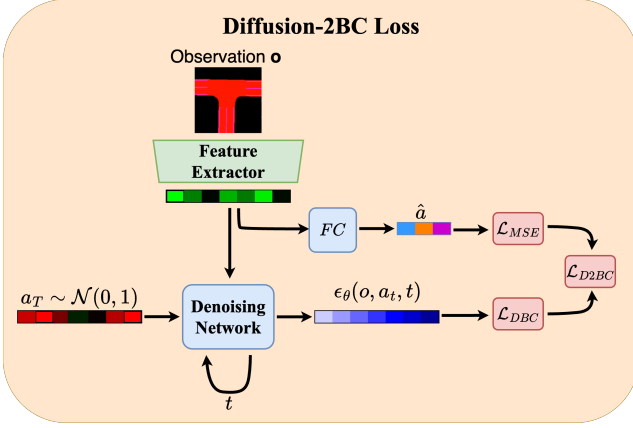


Fig. 1 Diffusion-2BC training architecture. The shared feature extractor processes the observation once. Its representation conditions the denoising branch used to compute $\mathcal{L}_{DBC}$ and feeds an auxiliary deterministic head used to compute $\mathcal{L}_{MSE}$. Their weighted sum updates the shared representation. The auxiliary head is discarded after training, and test-time actions are generated only by reverse diffusion

Table 2 Compact comparison of the evaluated policy architectures

Component	MSE-BC	Diffusion-BC	Diffusion-2BC
Visual encoder	Residual CNN	Residual CNN	Shared residual CNN
Executed head	CARLA: MLP 4608–2304–1024–64–2	Transformer denoiser	Transformer denoiser
Auxiliary head	–	–	CARLA: MLP 4608–2048–512–128–32–2
Inference	One deterministic pass	T reverse steps	T reverse steps

Table S2 provides the layer-by-layer specification.

4.4 Training and inference

Algorithm 1 summarizes vectorized minibatch training. A diffusion timestep and Gaussian noise are sampled independently for every action in the minibatch. The corrupted actions and observation features are processed by the denoising branch, while the same features are processed by the auxiliary regression branch. The two minibatch-averaged losses are combined before a single parameter update. During inference, the auxiliary prediction and regression loss are absent; consequently, Diffusion-2BC has the same test-time computational structure as Diffusion-BC.

4.5 Interpretation of the auxiliary loss

The auxiliary head should not be interpreted as a second policy used to arbitrate with the diffusion output. Its role is representation learning. The direct action target provides a short path from the expert controls to the visual features, while the denoising branch retains the distributional objective. This separation explains why a nonzero MSE weight need

Algorithm 1 Diffusion-2BC minibatch training

Require: Demonstrations $\mathcal{D}$; timesteps T ; diffusion schedule $\bar{\alpha}_{1:T}$; loss-weight schedule α_e

- 1: **for** each training epoch e **do**
- 2: Set $\alpha \leftarrow \alpha_e$ once for the current epoch
- 3: **for** each minibatch $(O, A) \sim \mathcal{D}$ of size B **do**
- 4: Sample $t_i \sim \mathcal{U}\{1, \dots, T\}$ and $\epsilon_i \sim \mathcal{N}(0, I)$ for $i = 1, \dots, B$
- 5: $A_t \leftarrow \sqrt{\bar{\alpha}_t} \odot A + \sqrt{1 - \bar{\alpha}_t} \odot \epsilon$
- 6: $H \leftarrow f_\theta(O)$
- 7: $\hat{\epsilon} \leftarrow \epsilon_\theta(H, A_t, t)$; $\hat{A} \leftarrow g_\theta(H)$
- 8: $L_{DBC} \leftarrow B^{-1} \sum_{i=1}^B \|\hat{\epsilon}_i - \epsilon_i\|_2^2$
- 9: $L_{MSE} \leftarrow B^{-1} \sum_{i=1}^B \|\hat{A}_i - A_i\|_2^2$
- 10: $L \leftarrow (1 - \alpha)L_{DBC} + \alpha L_{MSE}$
- 11: Update all trainable parameters using ∇L
- 12: **end for**
- 13: **end for**

not force deterministic inference: stochasticity is preserved because the deterministic prediction is not executed.

The formulation also clarifies two boundary cases. With $\alpha = 0$, the model reduces to Diffusion-BC. As α approaches one, denoising receives progressively less weight and the training objective approaches deterministic BC, although the remaining reverse model would be poorly trained. Useful operation therefore requires both terms. The experiments therefore evaluate two practical choices—a fixed weight and an exponentially decaying schedule—but leave a full schedule ablation to future work.

5 Experimental Methodology

5.1 Staged evaluation strategy

The experiments are organized according to the research question addressed by each environment:

1. **Controlled multimodal prediction:** Can diffusion-based BC and the hybrid objective reproduce several valid actions in the Claw environment?
2. **Route-conditioned urban navigation:** Can the methods learn stable trajectory following in CARLA from a small route dataset?
3. **Route-free urban navigation and transfer:** Can the methods drive through multiple intersections, express varied paths, and transfer from Town01 to unseen Town02?

This progression prevents the strongest CARLA claims from being inferred solely from simplified environments.

5.2 Claw environment

We adopt the Claw environment introduced in the original Diffusion-BC study [13] to evaluate multimodal behavior cloning. The environment uses 32×32 RGB top-down images containing one or more of five toy shapes. The

shapes are randomly translated, rotated, and scaled on a fixed background. The action is a two-dimensional point $a = (x, y)$, and an action is successful when it lies inside the mask of any object in the image. One successful point is sampled for each training image. Images containing multiple objects therefore admit several disconnected valid action regions, making the task a compact test of multimodal conditional prediction.

The training set contains 20,000 observation-action pairs. The models are trained for 150 epochs with learning rate 10^{-4} , cosine learning-rate decay, 512 hidden units, and batch size 32. Diffusion models use 50 denoising steps and a linear β schedule from 10^{-4} to 2×10^{-2} . Evaluation uses seven held-out observations and 300 sampled actions per observation. For each sampled action, the metric is the Euclidean distance to the nearest valid mask point; predictions inside a valid object have zero error. The reported score is the mean distance over all sampled actions and observations. The same protocol is repeated after reducing the training set to 90% of its original size.

The original Diffusion-BC paper presents representative input, mask, and sampled-action visualizations for this environment [13]; that figure is not reproduced here. The present study uses the same benchmark structure and reports a quantitative comparison of MSE-BC, Diffusion-BC, and the proposed Diffusion-2BC objective.

5.3 CARLA observations and controls

CARLA is an open-source urban-driving simulator with physics-based vehicle dynamics and configurable maps [22]. The experiments use a vehicle-centered bird’s-eye-view (BEV) representation rather than raw front cameras. The BEV is a 192×192 RGB image in which blue denotes the drivable road, magenta denotes lane boundaries, black denotes the non-drivable background, and yellow denotes the planned route when route conditioning is enabled. The two-dimensional action contains steering in $[-1, 1]$ and an acceleration command implemented as throttle in $[0, 1]$. Each BEV frame is converted to grayscale, stacked with the previous three frames to provide temporal information, and normalized, producing a $192 \times 192 \times 4$ tensor. The four frames are consecutive (frame stride 1), with no frame skipping. For the first three indices, the current frame is repeated four times to complete the input stack. The shared feature extractor uses convolutions, residual blocks, pooling, and a dense representation. MSE-BC maps this representation directly to the action, whereas Diffusion-BC and Diffusion-2BC condition a transformer-based denoising network. Fig. 2 shows examples of the two CARLA observation configurations.

BEV input reduces the perception burden and lets the study concentrate on policy learning. It should not be interpreted as eliminating the perception problem in a deployable system: a separate model or map-based process would be needed to construct BEV from real sensors. Previous work shows that cross-view transformations can learn this mapping [23, 24,

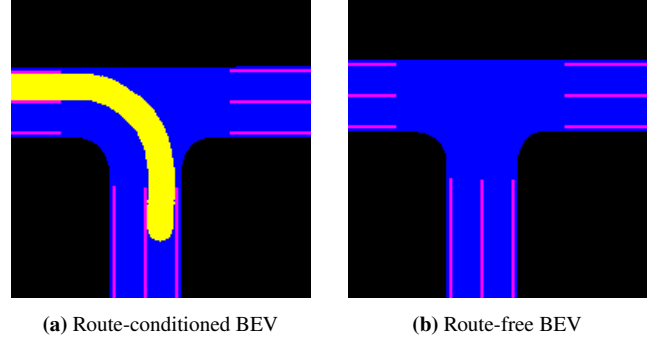


Fig. 2 Examples of the CARLA observations supplied to the policies. In (a), the yellow planned-route channel identifies the reference path, while blue and magenta encode the drivable road and lane boundaries. In (b), the route channel is omitted, so the policy must choose among locally valid road continuations. In both configurations, black denotes the non-drivable background.

25].

5.4 Route-conditioned CARLA protocol

A PID expert follows one CARLA Leaderboard route in Town01, producing approximately 1,900 observation-action pairs. The learning agent receives road, lane-boundary, and planned-route channels but not the expert’s dense control trajectory. Each architecture is trained three times. Evaluations occur every 50 epochs and use ten new random routes per model, with different lengths and curvatures. A rollout terminates at an infraction or after 3,000 steps. The reported metrics are distance traveled in meters and the fraction of the 3,000-step horizon completed.

Common hyperparameters are learning rate 10^{-4} , fixed learning rate, 128 hidden units in the feature extractor, and batch size 32. Optimization uses Adam with zero weight decay. Diffusion models use a linear β schedule from 10^{-4} to 2×10^{-2} , $T = 20$, embedding dimension 128, and a transformer denoising architecture. Diffusion-2BC uses fixed $\alpha = 0.3$ in this experiment.

5.5 Route-free multi-intersection protocol

The main multimodal dataset contains 34 PID-expert routes in Town01 and approximately 16,200 samples. It includes multiple intersection maneuvers, but the route channel is omitted from the agent’s input. Two models per architecture are trained through epoch 700 and evaluated at 10-epoch intervals. Each model is evaluated five times from the same initial location. No explicit random seed is set for training. The objectives are to travel as far as possible under the active termination criteria and to determine whether repeated rollouts take distinct valid paths.

Because no reference route is active, CARLA does not provide route length. Distance is accumulated from the

three-dimensional simulator coordinates:

$$\Delta x_t = x_t - x_{t-1}, \quad \Delta y_t = y_t - y_{t-1}, \quad (7)$$

$$d_t = \sqrt{\Delta x_t^2 + \Delta y_t^2}, \quad D = \sum_t d_t. \quad (8)$$

Because this simulator-specific quantity is used only for within-protocol comparison and does not correspond directly to a standardized physical distance under the present setup, it is reported as a unit of measurement (UoM) rather than meters. The wrong-lane termination rule is relaxed to allow wider turns, and red-light and stop-sign termination checks are disabled in the route-free protocol; the remaining configured infractions still terminate the rollout. The same trained policies are evaluated in Town01 and in Town02, which is not used for training. For each architecture, the five checkpoints with the highest mean distance in Town01 are fixed first and then transferred to Town02 for evaluation, without a new checkpoint search in Town02.

For convenient replication, Supplementary Table S3 consolidates the principal training and diffusion hyperparameters used in the CARLA experiments.

6 Results

6.1 Claw environment results

Table 4 reports the complete Claw results. With the full dataset, MSE-BC obtains a mean distance of 4.02, whereas Diffusion-BC and the two Diffusion-2BC configurations lie between 0.61 and 0.68. This pattern agrees with the benchmark structure: deterministic predictions tend to concentrate, whereas diffusion samples can cover separated valid regions. With 90% of the training data, fixed- α Diffusion-2BC reaches 0.63, compared with 0.70 for Diffusion-BC. These results show that the hybrid objective preserves the advantage of diffusion-based multimodal prediction in this controlled task; they are not used as evidence of autonomous-driving performance.

6.2 Route-conditioned CARLA navigation

Fig. 4 tracks distance and completion through training. Diffusion-2BC is the first method to complete every evaluation rollout without an infraction, reaching that behavior at epoch 150. Diffusion-BC reaches full completion at epoch 300. MSE-BC occasionally travels long distances but does not consistently complete all routes within the evaluated horizon and shows substantially higher dispersion.

At the checkpoints summarized in Table 5, Diffusion-2BC travels 1498.09 ± 74.81 m with completion 1.00 ± 0.00 . Diffusion-BC also reaches complete rollouts but covers 1339.85 ± 150.60 m. The apparent possibility of a distance above or below another method despite equal completion arises because the ten randomly generated routes vary and

because the vehicle speed affects distance covered within the 3,000-step horizon. The results indicate that the hybrid objective improves learning speed and stability in this route-following setting; they do not test multimodal route choice because the intended route is visible in the observation.

6.3 Route-free multi-intersection navigation in Town01

The multi-intersection experiment combines route-choice diversity with long-horizon closed-loop control. At its peak mean-distance checkpoint, MSE-BC travels 124.945 UoM on average. Its minimum, maximum, and mean are identical within each evaluation because repeated rollouts execute the same actions and overlap on the same trajectory. This deterministic consistency does not reproduce the route diversity present in the 34-route dataset.

Diffusion-BC reaches a lower peak mean of 51.873 UoM and exhibits a wide range of rollout lengths. At some evaluated checkpoints, one rollout travels only 4.501 UoM while another reaches 365.222 UoM. The method therefore generates stochastic trajectory variation, but this variation is not consistently associated with distinct intersection decisions or reliable progress.

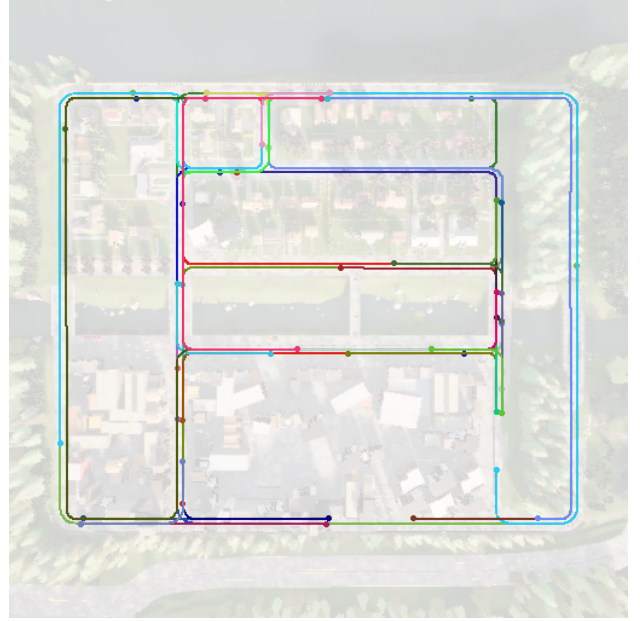
Diffusion-2BC attains the highest peak mean, 304.697 UoM, approximately 2.44 times the MSE-BC mean and 5.87 times the Diffusion-BC mean. Its range at the peak mean-distance checkpoint is 116.741–360.201 UoM. To avoid method-specific hand-selection, the cross-method trajectory comparison in Fig. 6 uses the same objective rule for all architectures: the checkpoint with the highest mean distance. These checkpoints are epoch 60 for MSE-BC, epoch 210 for Diffusion-BC, and epoch 330 for Diffusion-2BC. Because peak distance does not necessarily coincide with the clearest route diversity, epochs 340 and 640 of Diffusion-2BC are shown separately as additional qualitative examples; they are not used for the quantitative comparison in Table 6. The complete Town01 top-checkpoint histograms are provided as Supplementary Fig. S3.

6.4 Cross-map evaluation in Town02

The Town02 experiment evaluates policies trained only in Town01, without additional gradient updates, under changed road geometry. For each architecture, the five candidate checkpoints were first selected from Town01 according to mean-distance performance and were then evaluated in Town02 without retraining or a new sweep over training epochs. Fig. 5 reports Town02 performance for these Town01-preselected checkpoints from one trained instance of each architecture. Among these transferred candidates, MSE-BC reaches a highest observed mean of 70.563 UoM and has zero-width min–max bounds, confirming that all repeated rollouts follow the same path. Diffusion-BC reaches 165.717 UoM but remains highly inconsistent, with a 7.489–364.232 UoM range.



(a) Single route for route-conditioned training



(b) Thirty-four routes for route-free training

Fig. 3 CARLA demonstration datasets. Each colored line is a PID-expert trajectory recorded with the corresponding observations and controls. The route-conditioned dataset supplies a visible planned-route channel. The route-free dataset omits that channel and includes multiple intersection maneuvers, so similar BEV observations can be associated with different valid actions

Diffusion-2BC reaches the highest observed mean, 383.963 UoM, with a 150.107–429.379 UoM range.

Among the five Town01-preselected candidates evaluated in Town02, epochs 220, 230, and 680 yield the highest Town02 mean distances for MSE-BC, Diffusion-BC, and Diffusion-2BC, respectively. At epoch 680, the Diffusion-2BC rollouts predominantly share the same cyclic road structure. Epoch 50 is therefore shown only as an additional qualitative example because it provides the clearest Town02 instance of distinct route choices, including an inner and an outer route, while still demonstrating cross-map operation. This separation prevents a visually selected checkpoint from being presented as the quantitative optimum.

6.5 Computational cost

In the route-free CARLA experiment, MSE-BC requires approximately one minute per ten training epochs and five seconds per 100 inference steps. Diffusion-BC and Diffusion-2BC require approximately ten minutes per ten training epochs and 32 seconds per 100 inference steps. The proposed auxiliary branch does not increase test-time cost relative to Diffusion-BC because it is discarded after training. The reported implementation used Python 3.10 with PyTorch 1.13.1 and CUDA 11.7 runtime packages on an NVIDIA GeForce RTX 3060 GPU and an AMD Ryzen 5 5500 CPU. Both diffusion methods remain substantially slower than deterministic BC because each action is obtained through iterative denoising.

These measurements are implementation- and hardware-

dependent and are not normalized by parameter count or optimized sampling steps. They nevertheless reveal the practical trade-off: the highest-performing diffusion policy in these evaluations is not the fastest at inference. Deployment-oriented work should consider fewer denoising steps, distillation, action chunking, or a faster sampler, but those alternatives were not evaluated and are left for future work. Reaching a favorable checkpoint in fewer training epochs should therefore not be confused with lower execution latency.

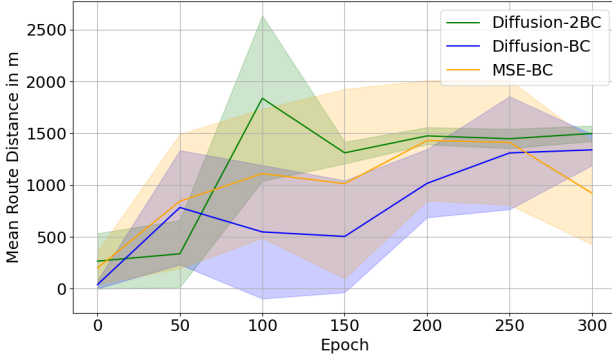
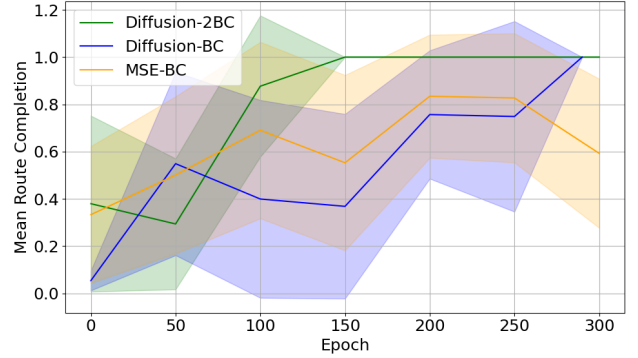
7 Discussion

The experiments support two complementary conclusions. First, the Claw benchmark confirms that diffusion-based objectives are better suited than deterministic MSE regression to conditional outputs with disconnected valid regions, and that the hybrid objective retains this advantage. Second, the auxiliary MSE objective improves closed-loop route progress in CARLA: Diffusion-2BC reaches full route-conditioned completion earlier than Diffusion-BC, produces the longest route-free trajectories at its peak checkpoints, and exhibits qualitative route variation at selected checkpoints in Town01 and Town02.

A plausible explanation is that direct action regression provides the shared visual encoder with a lower-variance supervised signal, while the denoising branch preserves a conditional action distribution. This interpretation is consistent with the results but is not a causal finding, because the study does not include gradient analysis, representation probes, or a systematic sweep of α and its schedule.

Table 3 Summary of experimental protocols

Experiment	Dataset	Models	Training	Evaluation	Main metrics	Generalization role
Claw environment	20,000 pairs; also 90% subset	MSE-BC, Diffusion-BC, Diffusion-2BC	Simplified supervised task	300 predictions	Mean mask-distance error	Controlled multimodality
Route-conditioned CARLA	~1,900 pairs from one Town01 route	All three	Three trainings; evaluations every 50 epochs	Ten new random routes; 3,000-step limit	Distance and relative completion	Novel routes in training map
Route-free CARLA	34 Town01 routes; ~16,200 pairs	All three	Two trainings; through epoch 700; evaluations every 10	Five repeated rollouts from same start	UoM distance and path diversity	Town01-selected checkpoints transferred to Town02

**(a)** Distance traveled before termination**(b)** Fraction of the 3,000-step horizon completed**Fig. 4** Route-conditioned CARLA evaluation during training. Each point combines three training runs and ten random evaluation routes per model. The curves show both progress and variability; an infraction terminates the rollout before the maximum horizon**Table 4** Results in the Claw environment. The score is the mean distance to the nearest valid mask region (lower is better), averaged over seven held-out observations with 300 sampled actions per observation

Method	Full dataset	90% dataset
MSE-BC	4.02	4.25
Diffusion-BC	0.68	0.70
Diffusion-2BC, exponential α	0.63	0.69
Diffusion-2BC, fixed $\alpha = 0.3$	0.61	0.63

The route-free evaluation also shows why diversity cannot be inferred from trajectory spread alone or assessed independently of task progress. The Town01 cross-method panels use the same peak mean-distance criterion for every architecture, whereas the Town02 panels are drawn from the fixed set of Town01-selected candidates. Separate Diffusion-2BC panels are explicitly labeled as qualitative examples because the visually clearest diversity checkpoint does not always coincide with the quantitative peak. Diffusion-BC generates stochastic variation but does not show a clear alternative

Table 5 Route-conditioned CARLA evaluation at the best completion-oriented checkpoints: epoch 200 for MSE-BC, epoch 300 for Diffusion-BC, and epoch 300 for Diffusion-2BC. Values aggregate three independently trained models, each evaluated on ten random routes; the reported $\pm$ values are standard deviations over the resulting 30 rollouts

Method	Distance (m)	Relative completion
MSE-BC	1430.05 $\pm$ 580.55	0.83 $\pm$ 0.26
Diffusion-BC	1339.85 $\pm$ 150.60	1.00 $\pm$ 0.00
Diffusion-2BC	1498.09 $\pm$ 74.81	1.00 $\pm$ 0.00

route at its peak checkpoint, whereas Diffusion-2BC exhibits distinct intersection choices at additional selected epochs; the route overlays provide complementary qualitative evidence. Future work should add route entropy, pairwise trajectory distance, action-distribution calibration, or counts of distinct intersection-decision sequences. Town02 demonstrates operation under changed road geometry using checkpoints preselected in Town01. Because the reported Town02 peak is the maximum among five preselected candidates rather than

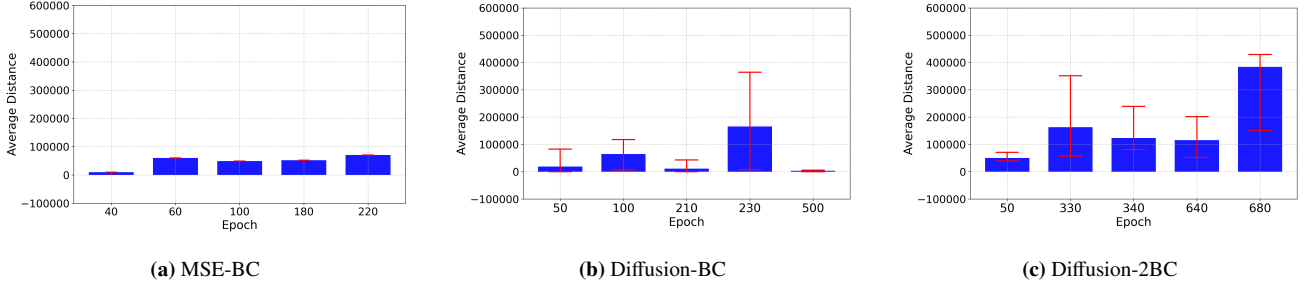


Fig. 5 Town02 cross-map performance for five checkpoints preselected from Town01 mean-distance performance, using one Town01-trained model per architecture. Bars show mean route-free distance and bounds show the minimum and maximum across repeated rollouts. MSE-BC repeats a single deterministic path, Diffusion-BC has large variability, and Diffusion-2BC achieves the highest peak mean distance

Table 6 Route-free multi-intersection distance results. All policies are trained in Town01, and Town02 is unseen during training. Town01 candidate checkpoints are selected by mean-distance performance; the Town02 row reports the highest Town02 mean among those five preselected candidates rather than a sweep over all training epochs.

Environment	Metric	MSE-BC	Diffusion-BC	Diffusion-2BC
Town01	Peak mean distance (UoM)	124.945	51.873	304.697
	Reported min-max at peak (UoM)	identical within runs	0.907–121.469	116.741–360.201
Town02	Highest mean in transferred set (UoM)	70.563	165.717	383.963
	Min-max at corresponding candidate (UoM)	identical within runs	7.489–364.232	150.107–429.379

the result of committing to a single checkpoint before transfer, it should still be interpreted as a descriptive cross-map result rather than a strict single-checkpoint generalization estimate.

While Diffusion-2BC demonstrates a marked reduction in trajectory failure relative to the baselines, residual variance remains an inherent feature of stochastic generative policies. In controlled traffic systems where specific paths are required, integrating high-level navigation goals (e.g., route conditioning or classifier-free guidance) provides an intuitive path to constrain the policy’s learned multimodal distribution to designated planner targets.

7.1 Scope of Study and Methodological Considerations

The experimental evaluation was designed to isolate and assess the core mechanisms of hybrid diffusion-regression learning in closed-loop vehicle control. To contextualize the findings within intelligent transportation systems, several methodological aspects define the operational scope of this study:

- **Evaluation Protocol and Cross-Map Transfer:** In the route-free autonomous navigation experiments, performance was characterized by tracking policy capability across training epochs to identify converged operational checkpoints. For the Town02 cross-map evaluation, candidate models were preselected exclusively based on their Town01 training metrics. Evaluating these preselected checkpoints in Town02 without additional fine-tuning or parameter updates provides a clear benchmark for zero-shot spatial transfer under altered road geometry and intersection

layouts.

- **Navigation Metrics:** The simulator-derived cumulative distance metric (UoM) was adopted to directly measure continuous closed-loop traversal endurance through arbitrary multi-lane intersections where fixed reference routes are absent. While standardized leaderboard driving scores incorporate composite infraction penalties for specific driving tasks, the coordinate-based distance metric provides an unconfounded, objective measure of an agent’s ability to remain within drivable boundaries over extended horizons.
- **Perceptual Representation and Environmental Complexity:** The use of Bird’s-Eye-View (BEV) semantic representations effectively decouples high-level policy decision-making from front-facing sensor perception artifacts, such as sensor noise, lens flare, or occlusion. Similarly, focusing the initial closed-loop validation on static urban networks allows for a controlled assessment of multimodal path selection, intersection negotiation, and trajectory continuity. This staged validation establishes baseline control stability prior to integrating dynamic surrounding traffic, adverse weather conditions, or heterogeneous vehicle-to-everything (V2X) interactions.
- **Comparative Scope:** The baseline comparisons were specifically chosen to isolate the architectural contribution of the proposed hybrid loss. By benchmarking Diffusion-2BC directly against its constituent components—pure deterministic regression (MSE-BC) and pure conditional diffusion (Diffusion-BC)—under identical visual encoders and network capacities, the results confirm that the auxiliary regression signal is the explicit driver of improved closed-loop consistency. Future extensions will examine how this

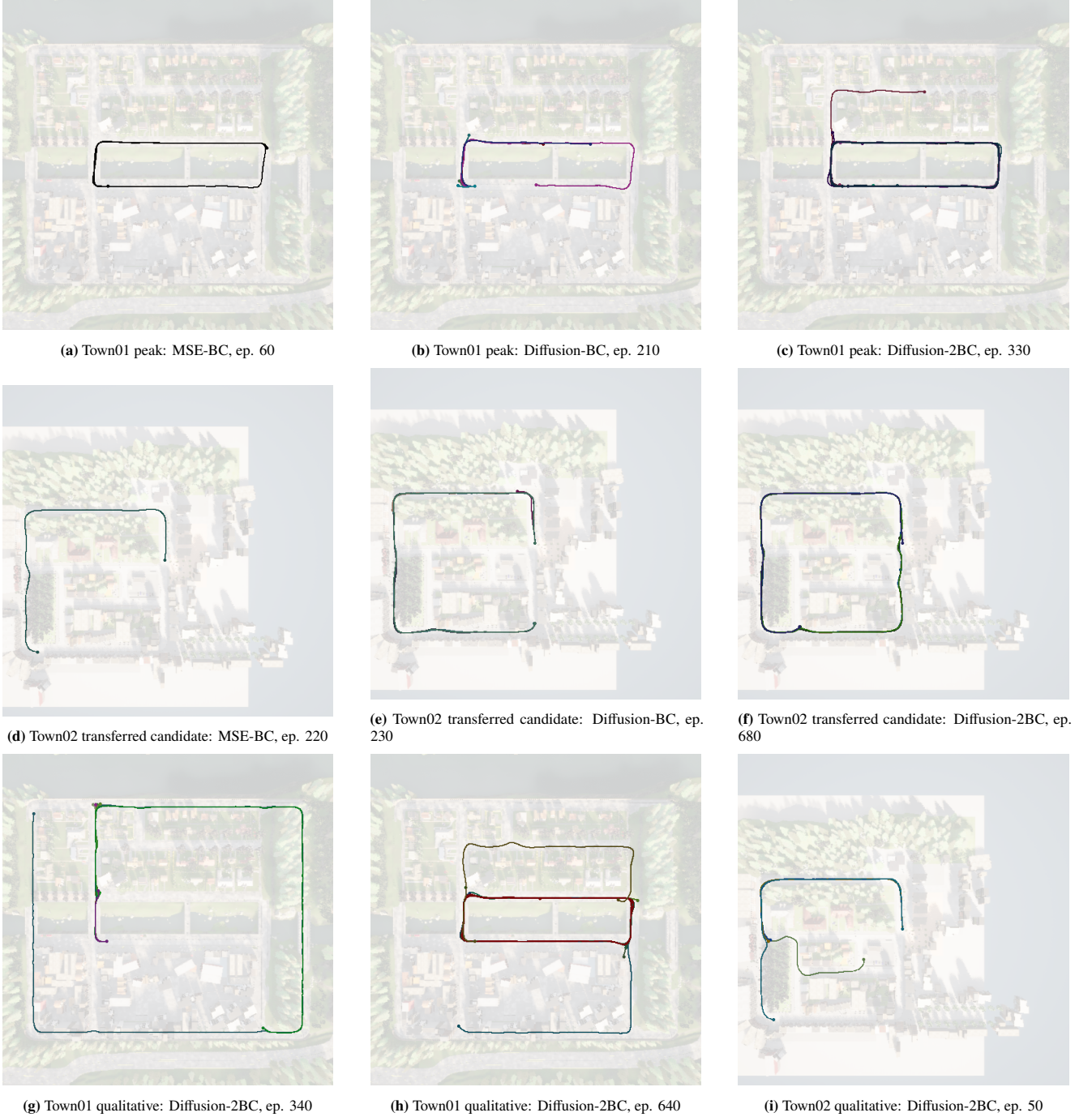


Fig. 6 Route-free rollouts; each panel overlays ten trajectories from the same initial point. The Town01 cross-method panels use the highest mean-distance checkpoint for each architecture (epochs 60, 210, and 330 for MSE-BC, Diffusion-BC, and Diffusion-2BC). The Town02 panels show epochs 220, 230, and 680 from the fixed set of five checkpoints preselected in Town01; these are the highest observed Town02 means within that transferred set, not checkpoints obtained by a new Town02 search. The last row contains additional Diffusion-2BC checkpoints selected only for qualitative inspection of route diversity. These qualitative panels are not used to compute or select the values in Table 6.

hybrid paradigm interfaces with alternative multimodal frameworks, such as energy-based or action-chunking architectures.

- **Experimental Reproducibility:** To reflect real-world learning conditions and ensure that reported behaviors

are robust rather than overfitted to specific initial conditions, the evaluation protocol aggregates multiple training instances and repeated stochastic rollouts. Detailed network specifications, training hyperparameter schedules, and CARLA environment settings are fully documented in

Sections 4 and 5 to support consistent implementation and replication.

8 Conclusions

This paper presented Diffusion-2BC, an offline behavior-cloning method that combines a diffusion denoising objective with an auxiliary deterministic regression loss. The auxiliary branch guides a shared visual feature extractor during training and is removed at test time, so the executed policy remains diffusion-based and capable of stochastic action generation.

The staged evaluation provides complementary evidence. In the Claw environment, the diffusion and hybrid objectives substantially reduce error relative to deterministic BC on observations with disconnected valid action regions. In route-conditioned CARLA navigation, Diffusion-2BC reaches infraction-free completion earlier and with lower variability. In the route-free experiment, it reaches the greatest peak mean distance, while selected checkpoints display qualitative route variation in Town01 and Town02.

The results justify a focused conclusion: under the evaluated datasets and baselines, an auxiliary regression signal improves the closed-loop reliability of diffusion behavior cloning while preserving multimodal prediction in the controlled Claw benchmark and qualitative route variation in CARLA. The results further indicate that the auxiliary deterministic signal can stabilize diffusion-based control in complex route-free driving scenarios while retaining qualitative evidence of behavioral diversity. This hybrid formulation provides a promising approach to learning-based urban vehicle control and motivates further investigation under more realistic traffic and perception conditions.

Future work will evaluate fixed, exponential, and cosine α schedules systematically; compare with energy-based and transformer-based multimodal policies; quantify route diversity; reduce the number of diffusion inference steps; add traffic, weather, traffic lights, and multi-agent interactions; and study command-guided selection of modes learned from unlabeled demonstrations. Future evaluations should also integrate real-time sensor-to-BEV perception pipelines and investigate hardware-accelerated diffusion inference. Beta-distribution diffusion noise is another possible direction because the driving controls are bounded [26].

Statements and Declarations

Funding

This work was partially funded by CNPq, Brazil (Grant No. 420148/2025-6).

Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Author contributions

Bruno Maciel Machado: Methodology, Software, Investigation, Data Curation, Visualization, and Writing—Original Draft.

Eric Aislan Antonelo: Conceptualization, Methodology, Formal Analysis, Supervision, Validation, Project Administration, Writing—Original Draft, and Writing—Review and Editing.

Data and code availability

The datasets, route definitions, trained models, and code supporting the findings of this study are available from the corresponding author upon reasonable request.

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Supplementary Material

Diffusion-2BC: Hybrid Diffusion and Regression Training for Offline Behavior Cloning in Autonomous Driving

Bruno Maciel Machado Eric Aislan Antonelo

Supplementary Note S1: Preliminary CarRacing experiment

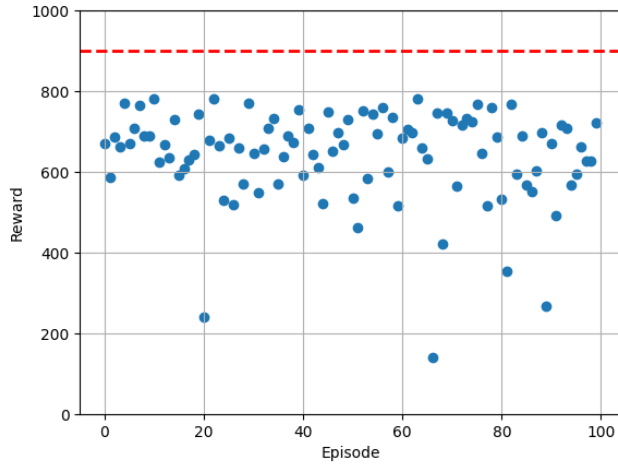
The main manuscript cites the CarRacing experiment as preliminary empirical motivation for combining diffusion-based action generation with a direct regression signal. The experiment predates Diffusion-2BC and compares only deterministic MSE-BC and Diffusion-BC; it is therefore not used as validation of the proposed method. This note records the protocol, reproduces the previously published reward distributions and aggregate statistics for completeness, and adds cumulative-reward visualizations that were not included in the preliminary conference article [14].

Environment and dataset. CarRacing-v2 provides 96×96 RGB observations and a continuous action vector containing steering, acceleration, and braking. Each observation was converted to grayscale, stacked with the three preceding frames, and normalized, resulting in a $96 \times 96 \times 4$ input. The fixed offline dataset contains approximately 15,000 observation–action pairs collected from 20 manually driven expert episodes on randomly generated tracks. Every demonstration obtained a reward above 900; the expert mean was 930.34 ± 7.46 .

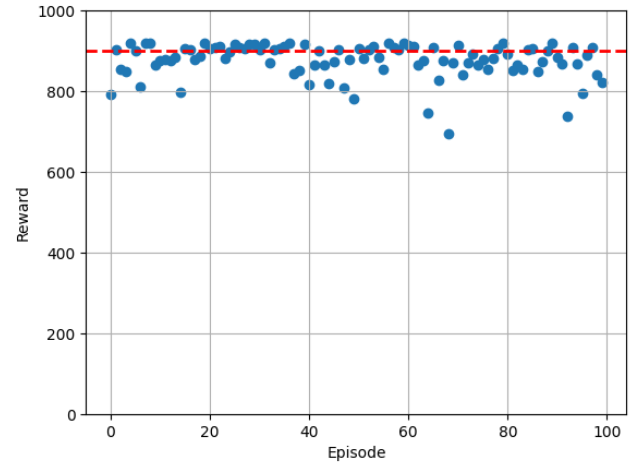
Training and evaluation. Both policies used the same convolutional feature extractor and were trained for 80 epochs with learning rate 10^{-4} , batch size 64, and 512 hidden feature units. Diffusion-BC used 50 diffusion timesteps, a linear β schedule from 10^{-4} to 2×10^{-2} , and a 128-dimensional embedding. Two independently trained models of each architecture were evaluated on 100 consecutive randomly generated tracks per model. A reward above 900 indicates expert-level completion under the environment’s progress-and-time reward.

Supplementary Table S1. Aggregate CarRacing performance across two independently trained models, each evaluated on 100 randomly generated tracks. These values were reported in the preliminary conference article.

Method	Mean reward	Standard deviation	Episodes above 900
MSE-BC	638.79	105.20	0%
Diffusion-BC	861.35	48.07	41%

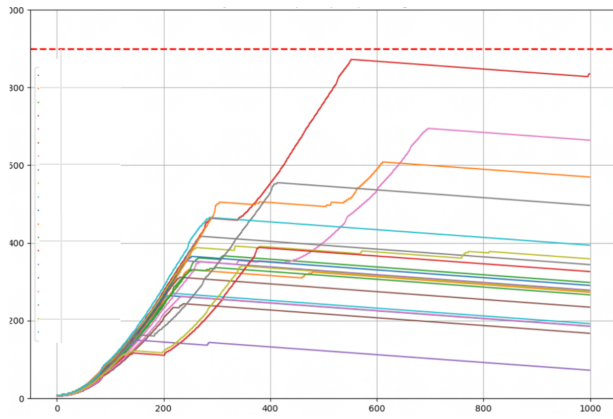


(a) MSE-BC: 642.73 ± 113.10 .

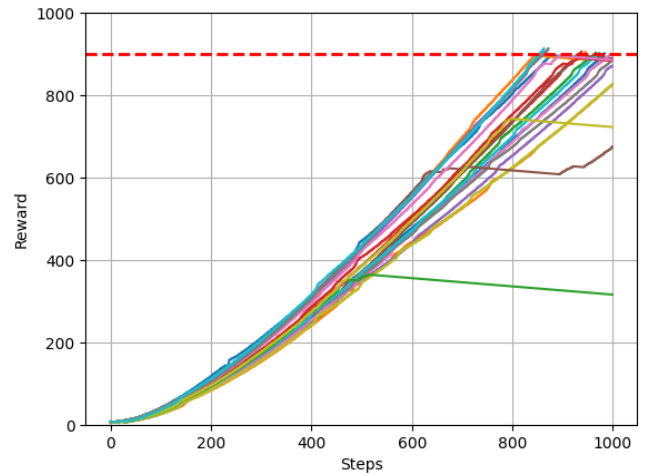


(b) Diffusion-BC: 876.39 ± 42.96 .

Supplementary Fig. S1. Reward distributions over 100 randomly generated evaluation tracks for one representative trained model of each architecture. The dashed red line marks reward 900. These distributions were previously reported in the conference article and are reproduced here to make the supplementary account self-contained.



(a) MSE-BC.



(b) Diffusion-BC.

Supplementary Fig. S2. Cumulative reward over environment steps for 20 representative evaluation episodes from one trained model of each architecture. These curves were not included in the preliminary conference article. The MSE-BC curves frequently plateau after missed track tiles, whereas most Diffusion-BC curves continue toward successful completion. The plots provide qualitative context for the aggregate statistics and are not evidence about Diffusion-2BC.

Supplementary Table S2: Layer-by-layer policy architectures

The table below expands the CARLA architecture summary in the main manuscript; the preliminary CarRacing configuration is documented separately in Supplementary Note S1. In CARLA, the four-frame 192×192 input produces a 4,608-dimensional flattened visual representation and the action dimension is $d_a = 2$.

Supplementary Table S2. Detailed CARLA neural-network components used by MSE-BC, Diffusion-BC, and Diffusion-2BC.

Component	Layer sequence and dimensions	Used by
Residual block 1	Two Conv2D layers, kernel 3×3 , stride 1, padding 1, 64 output channels; BatchNorm and GELU after each convolution; residual sum scaled by $1/\sqrt{2}$; MaxPool2D 2×2	All policies
Residual block 2	Two Conv2D layers with the same kernel, stride, padding, normalization, activation, and residual scaling; 128 output channels; MaxPool2D 2×2	All policies
Visual projection	AvgPool2D 8×8 , followed by flattening; resulting dimension depends on observation resolution	All policies
Deterministic policy head	Flattened visual vector $\rightarrow 2304 \rightarrow 1024 \rightarrow 64 \rightarrow d_a$ with ReLU after the hidden linear layers	MSE-BC
Observation embedding	4,608-dimensional flattened visual vector $\rightarrow 128 \rightarrow 128$ with LeakyReLU between the linear layers for CARLA	Diffusion-BC, Diffusion-2BC
Noisy-action embedding	$d_a \rightarrow 128 \rightarrow 128$ with LeakyReLU between the linear layers	Diffusion-BC, Diffusion-2BC
Timestep embedding	Scalar normalized timestep processed by a two-layer TimeSiren embedding with a sine activation, producing 128 features	Diffusion-BC, Diffusion-2BC
Transformer tokens	Observation, noisy-action, and timestep embeddings are separately projected from 128 to 64 features and supplied as three tokens with positional embeddings	Diffusion-BC, Diffusion-2BC
Transformer denoiser	Four encoder blocks; 16 attention heads; internal Q/K/V dimension $64 \times 16 = 1024$; feed-forward path $64 \rightarrow 256 \rightarrow 64$ with GELU; residual scaling and BatchNorm; no Transformer dropout	Diffusion-BC, Diffusion-2BC
Noise projection	Flatten three 64-dimensional output tokens and project $192 \rightarrow d_a$ to predict the action noise	Diffusion-BC, Diffusion-2BC
Auxiliary regression head	Flattened visual vector $\rightarrow 2048 \rightarrow 512 \rightarrow 128 \rightarrow 32 \rightarrow d_a$; BatchNorm, ReLU, and dropout $p = 0.3$ after each hidden linear layer	Diffusion-2BC training only
Inference	Direct deterministic action for MSE-BC; iterative reverse diffusion for Diffusion-BC and Diffusion-2BC; auxiliary regression head omitted for Diffusion-2BC	Method-dependent

Supplementary Table S3: CARLA experimental hyperparameters

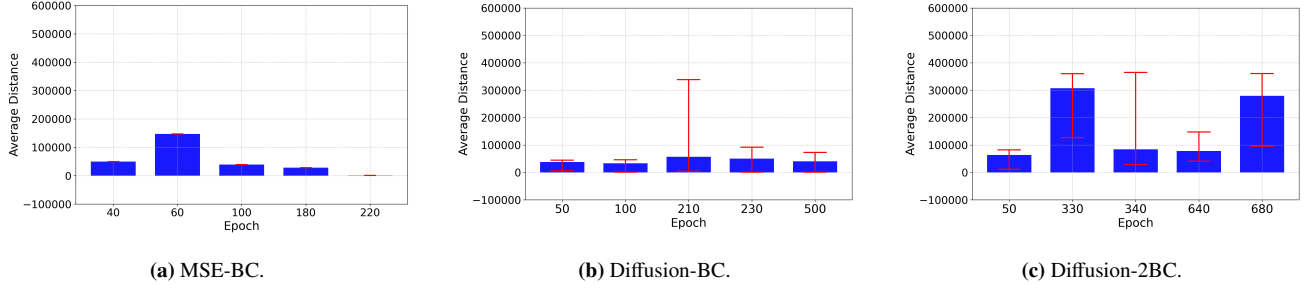
The main manuscript reports experiment-specific settings in the methodology. Supplementary Table S3 consolidates the principal training and diffusion hyperparameters used in the CARLA experiments for convenient replication.

Supplementary Table S3. Main training and diffusion hyperparameters for the CARLA experiments.

Parameter	Route-conditioned CARLA	Route-free CARLA
Learning rate	10^{-4}	10^{-4}
Learning-rate schedule	Fixed	Fixed
Feature hidden units	128	128
Batch size	32	32
Optimizer	Adam	Adam
Weight decay	0	0
Explicit training seed	Not set	Not set
Training epochs	300 for the primary comparison; Diffusion-2BC monitored to 750	700
Diffusion timesteps T	20	20
β limits	10^{-4} to 2×10^{-2}	10^{-4} to 2×10^{-2}
β schedule	Linear	Linear
Input embedding / denoiser	128 / Transformer	128 / Transformer
Context dropout	0	0
Guidance weight	0	0
Diffusion-2BC α	Fixed at 0.3	Exponential: $\alpha_e = \exp(-\lambda e)$, $\lambda = \ln(100)/E$; updated once per epoch
Evaluation frequency	Every 50 epochs	Every 10 epochs

Supplementary Fig. S3: Town01 top-checkpoint performance

The main manuscript prioritizes the Town02 cross-map histograms because performance under changed road geometry is more informative than training-map performance. For completeness, Supplementary Fig. S3 reports the corresponding Town01 training-map histograms for one trained model per architecture.



Supplementary Fig. S3. Town01 route-free performance for the five highest mean-distance checkpoints of one trained model per architecture. Bars show mean distance and bounds show the minimum and maximum across repeated rollouts.